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Wireshark Network Traffic Capture and Analysis

1. Objective

The objective of this assignment is to capture and analyze network traffic using Wireshark. The capture was performed on the active `eth0` network interface while browsing several websites. The captured traffic was analyzed to identify protocols, DNS queries, top communicating hosts, and any unusual network activity.

2. Tools Used

- Kali Linux
- Wireshark
- `eth0` network interface

3. Capture Procedure

1. Wireshark was opened in Kali Linux.
2. The active `eth0` network interface was selected.
3. Network traffic was captured while browsing several websites.
4. After completing the browsing session, the packet capture was stopped.
5. The capture was saved as `Sabbir_Wireshark_Capture.pcapng`.
6. The captured packets were analyzed using Wireshark Statistics and display filters.

4. Protocol Analysis

The Protocol Hierarchy Statistics showed that the capture contained several network protocols.

The capture contained approximately 11,175 packets. IPv4 accounted for approximately 99.7% of the packets. UDP represented approximately 75.0% of the packets, while QUIC accounted for approximately 73.2%. TCP represented approximately 24.6% and TLS represented approximately 10.2%. DNS traffic accounted for approximately 1.8% of the packets, and ARP accounted for approximately 0.3%.

The large amount of QUIC and TLS traffic is consistent with modern web browsing, where HTTPS and HTTP/3-based connections are commonly used.



5. DNS Query Analysis

DNS traffic was analyzed using the Wireshark display filter:

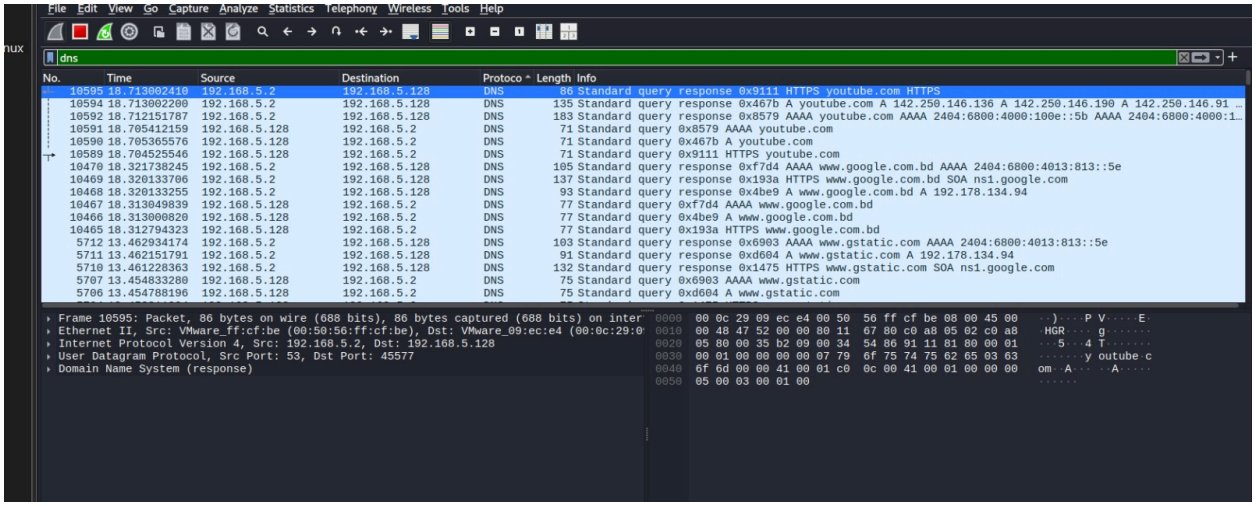
`dns`

The capture contained DNS queries and responses between the local machine (`192.168.5.128`) and the DNS server (`192.168.5.2`).

Several domain names were observed, including:

- `youtube.com`
- `www.google.com.bd`
- `www.gstatic.com`

Both A and AAAA DNS records were observed. A records provide IPv4 addresses, while AAAA records provide IPv6 addresses.



6. Top Talkers / Endpoint Analysis

The IPv4 Endpoints statistics were used to identify hosts involved in the captured traffic.

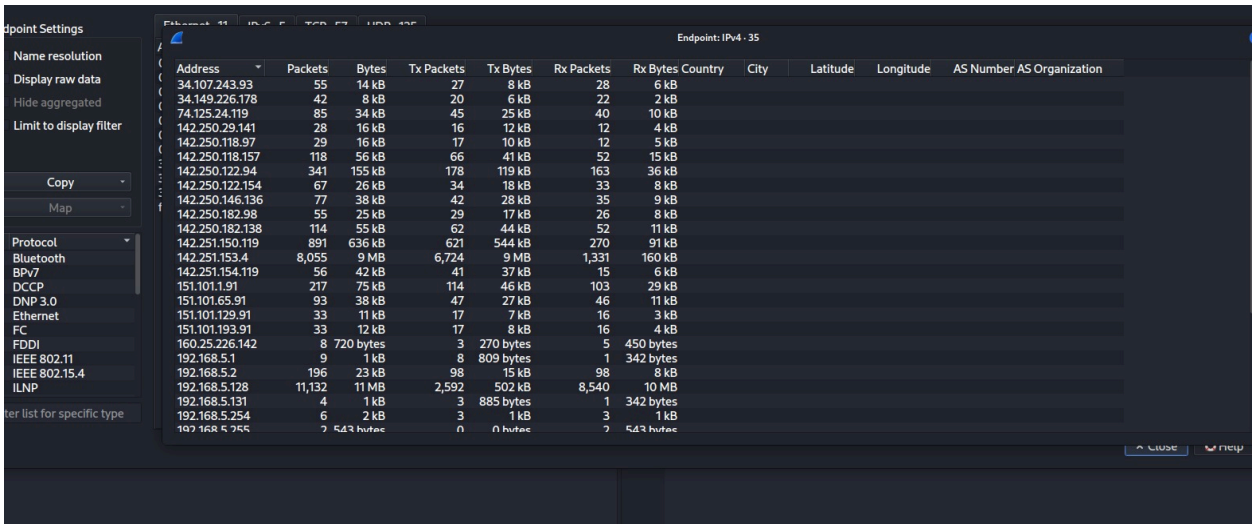
The local machine was identified as:

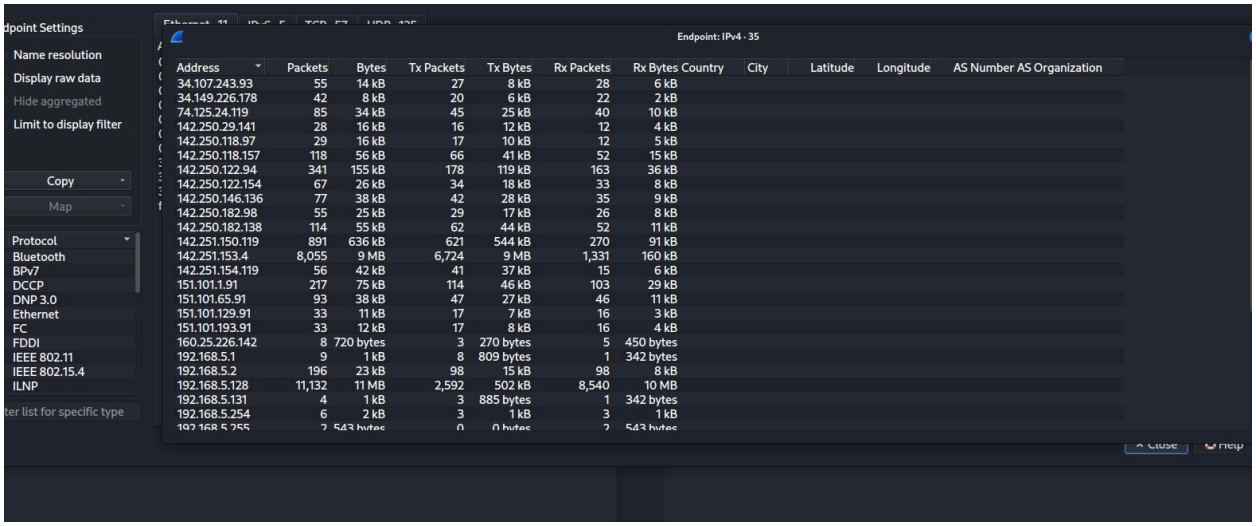
192.168.5.128

It generated/received approximately 11,132 packets and around 11 MB of traffic.

Among the external hosts, 142.251.153.4 was one of the highest-traffic endpoints, with approximately 8,055 packets and around 9 MB of traffic.

Other external IP addresses also communicated with the local machine during the browsing session.





7. Unusual Activity Analysis

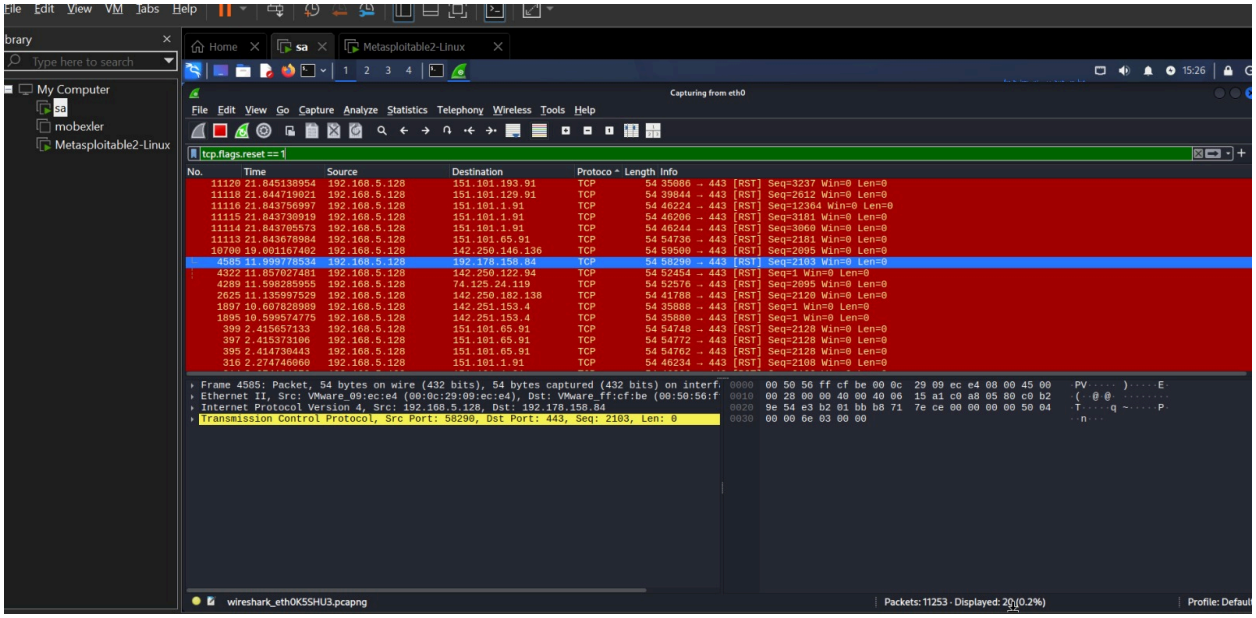
TCP Reset (RST) packets were analyzed using the Wireshark display filter:

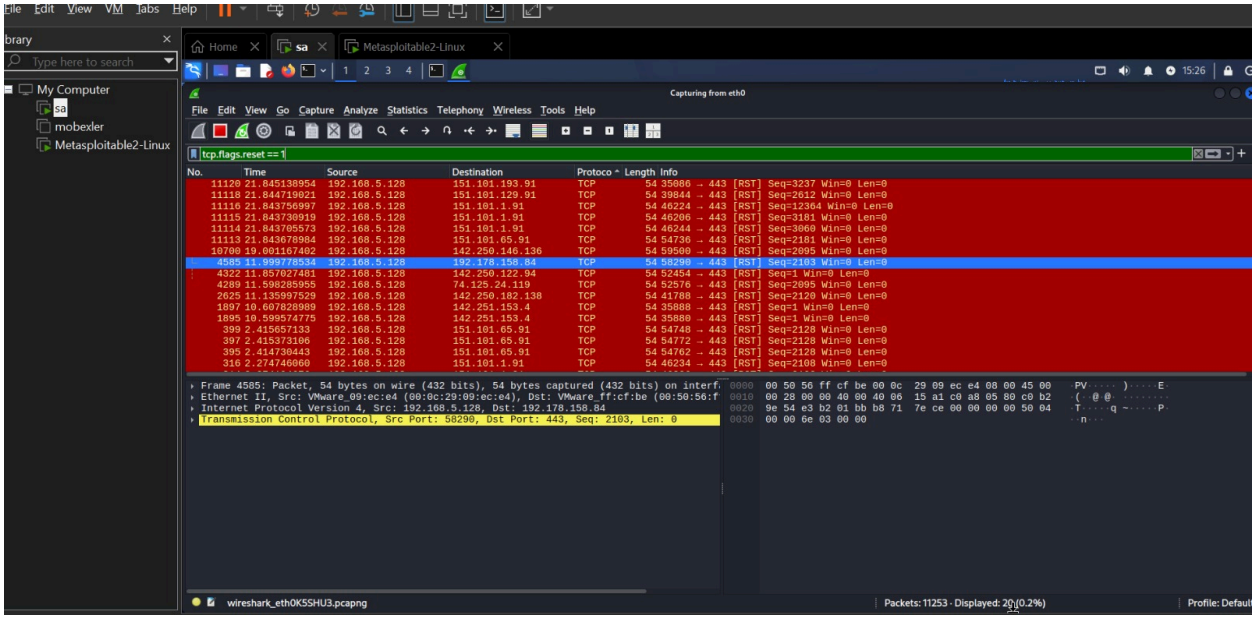
```
tcp.flags.reset == 1
```

Multiple TCP RST packets were observed, mainly involving connections to port 443.

A TCP RST indicates that a TCP connection was reset or terminated abruptly. The presence of RST packets alone does not prove malicious activity. They can occur for normal reasons such as closed connections, application behavior, or network interruptions.

Based on the captured traffic, no confirmed malicious activity was identified solely from these packets.





8. Findings

The analysis showed that the captured traffic was mainly normal web-browsing traffic. The major protocols observed were UDP/QUIC, TCP, TLS, DNS, and ARP. DNS queries for websites such as YouTube and Google were observed. Several external endpoints communicated with the local machine, with some generating significantly more traffic than others.

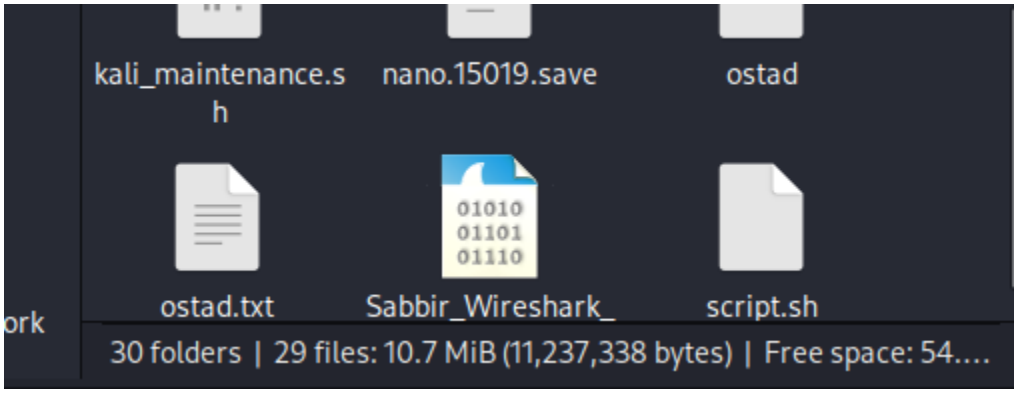
TCP RST packets were also observed, but these packets alone were not sufficient to classify the activity as malicious.

9. Conclusion

Wireshark was successfully used to capture and analyze network traffic. The analysis identified the protocols used, DNS queries, major communicating endpoints, and TCP reset activity. Overall, the captured traffic appeared consistent with normal web browsing, and no confirmed malicious activity was identified from the analyzed packets.

10. Submission Files

- Wireshark capture file: `Sabbir_Wireshark_Capture.pcapng`



`Drive Link:`

https://drive.google.com/file/d/16fe_LAbuZimlfl3_0VMGAo9wQ6-Dwsxa/view?usp=sharing

Conclusion:

The Wireshark capture was successfully performed and analyzed using the `eth0` network interface. The analysis identified several protocols, including IPv4, UDP, QUIC, TCP, TLS, DNS, and ARP. DNS queries for websites such as YouTube and Google were observed, and the IPv4 endpoint analysis identified the major communicating hosts. TCP RST packets were also detected, but they were not sufficient to confirm any malicious activity. Overall, the captured traffic appeared to be consistent with normal web browsing, and no confirmed security threat was identified during the analysis.